CHAPTER 2

Complex numbers: the geometric point of view

Complex plane, modulus, argument, trigonometric form

Objectives. Represent a complex number by a point or a vector; determine a modulus and an argument; move from the algebraic form to the trigonometric form and back; interpret a modulus as a distance.

Prerequisites. Chapter 1; coordinates in the plane, distance between two points; unit circle, radians, standard values; scalar product and determinant of two vectors.

Fluency

Fluency · No calculator, five minutes.

1. Compute the distance between $A(1,-2)$ and $B(4,2)$.
2. Give $\cos\frac{3\pi}{4}$ and $\sin\frac{3\pi}{4}$.
3. Determine the real number $\theta \in (-\pi,\pi]$ such that $\cos\theta = -\frac{1}{2}$ and $\sin\theta = \frac{\sqrt{3}}{2}$.
4. Give the principal measure of the angle $\frac{17\pi}{6}$.
5. Are the vectors $\vec{u}\begin{pmatrix}2\\1\end{pmatrix}$ and $\vec{v}\begin{pmatrix}-1\\2\end{pmatrix}$ orthogonal?
6. Write $(2+i)(1+i)$ in algebraic form.

Solution · Fluency.

1. $AB = \sqrt{3^2+4^2} = 5$.
2. $\cos\frac{3\pi}{4} = -\frac{\sqrt{2}}{2}$ and $\sin\frac{3\pi}{4} = \frac{\sqrt{2}}{2}$.
3. $\theta = \frac{2\pi}{3}$.
4. $\frac{17\pi}{6} - 2\pi = \frac{5\pi}{6}$, which lies in $(-\pi,\pi]$.

5. $\vec{u} \cdot \vec{v} = -2 + 2 = 0$: yes.

6. $2 + 2i + i - 1 = 1 + 3i$.

Introductory activity

Activity 1 · Is multiplying the same as turning? 35 min . The plane has an orthonormal frame $(O, \vec{u}, \vec{v})$. To a complex number $x + iy$ we associate the point with coordinates (x, y).

1. Plot the points A, B, C associated with $a = 2 + i$, $b = i\,a$ and $c = (1 + i)\,a$. Give b and c in algebraic form.
2. Compute the distances OA, OB and OC.
3. Compute the scalar product $\overrightarrow{OA} \cdot \overrightarrow{OB}$. What can be said about the lines (OA) and (OB)?
4. Compute $\cos\left(\overrightarrow{OA}, \overrightarrow{OC}\right)$ with the scalar product, then the sign of the determinant $\det\left(\overrightarrow{OA}, \overrightarrow{OC}\right)$. Deduce a measure of the oriented angle $\left(\overrightarrow{OA}, \overrightarrow{OC}\right)$.
5. Start again with $a = -1 + 3i$ (or a point of your choice). State a conjecture about the geometric effect of a multiplication by i, then by $1 + i$.
6. Which quantities attached to a complex number would have to be defined to state this conjecture in general?

Observing is not proving: the properties conjectured here are established in the course.

Solution · Activity.

1. $b = i(2 + i) = -1 + 2i$ and $c = (1 + i)(2 + i) = 1 + 3i$.
2. $OA = \sqrt{5}$, $OB = \sqrt{5}$, $OC = \sqrt{10}$.
3. $\overrightarrow{OA} \cdot \overrightarrow{OB} = 2 \times (-1) + 1 \times 2 = 0$: the lines are perpendicular, and $OB = OA$.
4. $\cos\left(\overrightarrow{OA}, \overrightarrow{OC}\right) = \frac{2\times1+1\times3}{\sqrt{5}\sqrt{10}} = \frac{5}{5\sqrt{2}} = \frac{\sqrt{2}}{2}$ and $\det\left(\overrightarrow{OA}, \overrightarrow{OC}\right) = 2 \times 3 - 1 \times 1 = 5 > 0$. A measure is $\frac{\pi}{4}$.
5. With $a = -1 + 3i$: $ia = -3 - i$ and $(1 + i)a = -4 + 2i$. Again we observe $OB = OA$ with a direct quarter turn, and $OC = \sqrt{2}\,OA$ with a direct eighth of a turn. Conjecture: multiplying by i turns through an angle $\frac{\pi}{2}$ about O; multiplying by $1 + i$ turns through $\frac{\pi}{4}$ and multiplies the distances to O by $\sqrt{2}$.
6. A length (the distance to O) and an angle (measured from $\vec{u}$): these are the modulus and the argument. The conjecture will be proved in chapter 3.

Teacher note · Running the activity. Let the students choose their second point and compare the results. Clearly separate the observation on examples from the proof, which will come with the formula $\arg(zz') = \arg z + \arg z'$ (chapter 3). Dynamic geometry software lets one move A and follow B and C.

Course notes

Complex plane

Definition · Image, affix. The plane has a direct orthonormal frame $(O, \vec{u}, \vec{v})$. To the complex number $z = a+ib$ (a, b real) we associate the point $M(a, b)$, called the *image* of z, and the vector $\vec{w}\begin{pmatrix}a\\b\end{pmatrix}$. We say that z is the *affix* of M and of $\vec{w}$. The plane with this correspondence is called the *complex plane*.

Property · Affixes and operations. If A and B have affixes z_A and z_B, the vector $\overrightarrow{AB}$ has affix $z_B - z_A$ and the midpoint of $[AB]$ has affix $\frac{z_A+z_B}{2}$. The affix of $\vec{w} + \vec{w'}$ is the sum of the affixes, that of $k\vec{w}$ (k real) is k times the affix of $\vec{w}$.

The image of $\overline{z}$ is the reflection of the image of z in the horizontal axis; the image of $-z$ is its reflection in O.

Modulus

Definition · Modulus. The *modulus* of $z = a + ib$ is the nonnegative real number $|z| = \sqrt{a^2 + b^2}$. If M is the image of z, $|z| = OM$. For a real number, the modulus is the absolute value.

Property · Modulus and conjugate. For every complex number z, $|z|^2 = z\overline{z}$, and $|\overline{z}| = |-z| = |z|$. If A and B have affixes z_A and z_B, then $AB = |z_B - z_A|$.

Proof. With $z = a + ib$, $z\overline{z} = a^2 + b^2 = |z|^2$ (chapter 1). The distance AB is the norm of the vector $\overrightarrow{AB}$, whose affix is $z_B - z_A$. □

Property · Modulus of a product, of a power, of a quotient. For all complex numbers z, z' and every natural number n:

$$|zz'| = |z|\,|z'|, \qquad |z^n| = |z|^n, \qquad \text{and, if } z' \neq 0, \quad \left|\frac{1}{z'}\right| = \frac{1}{|z'|}, \quad \left|\frac{z}{z'}\right| = \frac{|z|}{|z'|}.$$

Proof. *Product.* $|zz'|^2 = zz'\,\overline{zz'} = z\overline{z}\,z'\overline{z'} = |z|^2\,|z'|^2$. Moduli are nonnegative, so $|zz'| = |z|\,|z'|$.
Power. By induction: $|z^0| = 1 = |z|^0$, and if $|z^n| = |z|^n$, then $|z^{n+1}| = |z^n|\,|z| = |z|^{n+1}$.
Inverse and quotient. $|z'| \times \left|\frac{1}{z'}\right| = \left|z' \times \frac{1}{z'}\right| = 1$, then $\left|\frac{z}{z'}\right| = |z|\left|\frac{1}{z'}\right|$. □

Property · Triangle inequality, taken for granted here. For all complex numbers z and z', $|z + z'| \leq |z| + |z'|$. The problem of the chapter gives a proof and the case of equality.

Definition · The set $\mathbb{U}$. We write $\mathbb{U}$ for the set of complex numbers of modulus 1. Their images form the circle with centre O and radius 1.

Property · Stability of $\mathbb{U}$. If z and z' belong to $\mathbb{U}$, then $zz' \in \mathbb{U}$ and $\frac{1}{z} \in \mathbb{U}$; moreover $\frac{1}{z} = \overline{z}$.

Proof. $|zz'| = 1 \times 1 = 1$ and $\left|\frac{1}{z}\right| = \frac{1}{1} = 1$. Finally $z\overline{z} = |z|^2 = 1$, so $\overline{z} = \frac{1}{z}$. □

Method · Reading a set of points. Let A and B have affixes a and b, and M have affix z. The equality $|z-a| = r$ $(r > 0)$ means $AM = r$: M describes the circle with centre A and radius r. The equality $|z-a| = |z-b|$ $(a \neq b)$ means $AM = BM$: M describes the perpendicular bisector of $[AB]$.

Argument and trigonometric form

Definition · Argument. Let z be a nonzero complex number with image M. An *argument* of z is any measure, in radians, of the oriented angle $\left(\vec{u}, \overrightarrow{OM}\right)$. We write $\arg z$ for one of them; the others differ from it by a multiple of 2π, which we write $\arg z \equiv \theta \pmod{2\pi}$. The number 0 has no argument.

Property · Trigonometric form. Every nonzero complex number $z = a + ib$ can be written $z = r(\cos\theta + i\sin\theta)$ with $r = |z| > 0$ and $\theta \equiv \arg z \pmod{2\pi}$; then

$$\cos\theta = \frac{a}{r}, \qquad \sin\theta = \frac{b}{r}.$$

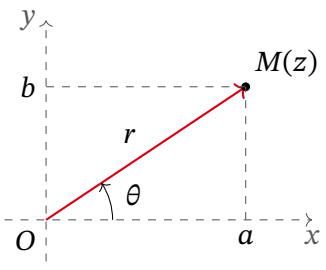
Conversely, if $z = r(\cos\theta + i\sin\theta)$ with $r > 0$, then $|z| = r$ and θ is an argument of z. Two trigonometric forms $r(\cos\theta + i\sin\theta)$ and $r'(\cos\theta' + i\sin\theta')$ are equal if and only if $r = r'$ and $\theta \equiv \theta' \pmod{2\pi}$.

Property · First arguments. For $z \neq 0$: $\arg\overline{z} \equiv -\arg z$ and $\arg(-z) \equiv \arg z + \pi \pmod{2\pi}$. A positive real number has argument 0, a negative real number π; i has argument $\frac{\pi}{2}$.

Method · Finding a trigonometric form. Compute $r = |z|$, then $\frac{a}{r}$ and $\frac{b}{r}$, and recognise the cosine and sine of a standard angle.

Example 1 · Four trigonometric forms. $1 + i\sqrt{3}$: $r = 2$, $\cos\theta = \frac{1}{2}$, $\sin\theta = \frac{\sqrt{3}}{2}$, so $1 + i\sqrt{3} = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$.
$-1 + i = \sqrt{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$; $-3 = 3(\cos\pi + i\sin\pi)$; $-2i = 2\left(\cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)\right)$.

Watch out. $-2\left(\cos\frac{\pi}{5} + i\sin\frac{\pi}{5}\right)$ is not a trigonometric form, because $-2 < 0$. We write $-2\left(\cos\frac{\pi}{5} + i\sin\frac{\pi}{5}\right) = 2\left(\cos\frac{6\pi}{5} + i\sin\frac{6\pi}{5}\right)$.



Modulus and argument of $z = a + ib$.

Key point · Key points. $|z| = OM$, $|z_B - z_A| = AB$, $|z|^2 = z\bar{z}$; the modulus of a product is the product of the moduli; $z = r(\cos\theta + i\sin\theta)$ with $r > 0$; an argument is only defined up to a multiple of 2π.

Exercises

Exercise 2.1 · Modulus and argument. Determine the modulus and an argument of: $1-i$; $-\sqrt{3}+i$; -5; $4i$.

Solution. $1-i$: $\sqrt{2}$, $-\frac{\pi}{4}$. $-\sqrt{3}+i$: 2, $\frac{5\pi}{6}$ (cosine $-\frac{\sqrt{3}}{2}$, sine $\frac{1}{2}$). -5: 5, π. $4i$: 4, $\frac{\pi}{2}$.

Exercise 2.2 · From the trigonometric form to the algebraic form. Write in algebraic form: $2\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)$; $\sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right)+i\sin\left(-\frac{\pi}{4}\right)\right)$; $6\left(\cos\frac{5\pi}{6}+i\sin\frac{5\pi}{6}\right)$.

Solution. $2\left(-\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)=-1+i\sqrt{3}$; $\sqrt{2}\left(\frac{\sqrt{2}}{2}-i\frac{\sqrt{2}}{2}\right)=1-i$; $6\left(-\frac{\sqrt{3}}{2}+\frac{i}{2}\right)=-3\sqrt{3}+3i$.

Exercise 2.3 · Computing a modulus without expanding. Compute the modulus of $(1+i)^6\left(\sqrt{3}-i\right)$, then that of $\dfrac{3+4i}{1-2i}$.

Solution. $\left|(1+i)^6(\sqrt{3}-i)\right|=\left(\sqrt{2}\right)^6\times 2=8\times 2=16$. Next $\left|\dfrac{3+4i}{1-2i}\right|=\dfrac{5}{\sqrt{5}}=\sqrt{5}$.

Exercise 2.4 · Sets of points. Determine and draw the set of points M with affix z such that:

1. $|z-2+i|=3$;
2. $|z-1|=|z+i|$;
3. $|\overline{z}+2i|=1$.

Solution.

1. $|z-(2-i)|=3$: circle with centre the point of affix $2-i$ and radius 3.
2. With $A(1)$ and $B(-i)$, $AM=BM$: perpendicular bisector of $[AB]$. Setting $z=x+iy$: $(x-1)^2+y^2=x^2+(y+1)^2$, that is $y=-x$.
3. $\overline{z}+2i=\overline{z-2i}$, so $|\overline{z}+2i|=|z-2i|$: circle with centre the point of affix $2i$ and radius 1.

Exercise 2.5 · Nature of a triangle. Consider the points A, B, C with affixes $a=1+i$, $b=3+2i$ and $c=2+4i$. Compute AB, BC and AC, then determine the nature of the triangle ABC.

Solution. $AB=|2+i|=\sqrt{5}$, $BC=|-1+2i|=\sqrt{5}$, $AC=|1+3i|=\sqrt{10}$. Since $AB=BC$ and $AB^2+BC^2=10=AC^2$, the triangle is right-angled and isosceles at B (converse of Pythagoras' theorem).

Exercise 2.6 · Numbers of modulus 1.

1. Prove that if $|z| = 1$, then $z + \dfrac{1}{z}$ is real.
2. Let z and z' be two elements of $\mathbb{U}$ such that $zz' \neq -1$. Prove that $\dfrac{z + z'}{1 + zz'}$ is real.

Solution.

1. $\dfrac{1}{z} = \overline{z}$, so $z + \dfrac{1}{z} = z + \overline{z} = 2\,\mathrm{Re}(z)$, which is real.
2. The conjugate is $\dfrac{\overline{z} + \overline{z'}}{1 + \overline{z}\,\overline{z'}} = \dfrac{\frac{1}{z} + \frac{1}{z'}}{1 + \frac{1}{zz'}}$. Multiplying numerator and denominator by zz' gives $\dfrac{z' + z}{zz' + 1}$: the number equals its conjugate, so it is real.

Exercise 2.7 · A sequence of complex numbers. Let $z_0 = 4$ and, for every natural number n, $z_{n+1} = \dfrac{1 + i}{2}\, z_n$.

1. Compute z_1, z_2 and plot the corresponding points.
2. Prove that $(|z_n|)$ is a geometric sequence; give its common ratio and its general term.
3. Determine the limit of $|z_n|$. Interpret geometrically.
4. Determine the smallest integer n such that $|z_n| < 0.01$.

Solution.

1. $z_1 = 2 + 2i$ and $z_2 = \dfrac{(1+i)(2+2i)}{2} = (1 + i)^2 = 2i$.
2. $|z_{n+1}| = \left|\dfrac{1+i}{2}\right| |z_n| = \dfrac{\sqrt{2}}{2}\,|z_n|$: a geometric sequence with common ratio $\dfrac{\sqrt{2}}{2}$ and first term 4, so $|z_n| = 4\left(\dfrac{\sqrt{2}}{2}\right)^n$.
3. $0 < \dfrac{\sqrt{2}}{2} < 1$, so $|z_n| \to 0$: the points get closer to O while turning (by an eighth of a turn at each step, as chapter 3 will justify).
4. $4\left(\dfrac{\sqrt{2}}{2}\right)^n < 0.01$ is equivalent to $n \ln \dfrac{\sqrt{2}}{2} < \ln 0.0025$, that is $n > \dfrac{\ln 0.0025}{\ln(\sqrt{2}/2)} \simeq 17.29$. The smallest integer is 18: $|z_{17}| \simeq 0.0110$ and $|z_{18}| = \dfrac{4}{512} \simeq 0.0078$.

Problem

Problem · Triangle inequality and the case of equality.

1. Let z be a complex number. Prove that $\mathrm{Re}(z) \leq |z|$, with equality if and only if z is a nonnegative real number.
2. Let z and z' be two complex numbers. Prove that $|z+z'|^2 = |z|^2 + |z'|^2 + 2\,\mathrm{Re}\left(z\overline{z'}\right)$.
3. Deduce the triangle inequality $|z+z'| \leq |z| + |z'|$.
4. Assume $z' \neq 0$. Prove that equality $|z+z'| = |z| + |z'|$ holds if and only if there is a real number $\lambda \geq 0$ such that $z = \lambda z'$. Interpret with the images M and M' of z and z'.
5. Application. Consider the points M with affix z such that $|z-3-4i| \leq 2$.
   (a) Prove that for these points, $3 \leq |z| \leq 7$.
   (b) Determine the affixes of the points for which $|z| = 7$, then $|z| = 3$.

Solution · Problem.

1. With $z = a + ib$: $\mathrm{Re}(z) = a \leq |a| = \sqrt{a^2} \leq \sqrt{a^2+b^2} = |z|$. Equality forces $b = 0$ and $a = |a|$, so z is a nonnegative real number; the converse is immediate.
2. $|z+z'|^2 = (z+z')\left(\overline{z}+\overline{z'}\right) = z\overline{z} + z'\overline{z'} + z\overline{z'} + \overline{z}z'$. Now $\overline{z}z' = \overline{z\overline{z'}}$, so $z\overline{z'} + \overline{z}z' = 2\,\mathrm{Re}\left(z\overline{z'}\right)$.
3. By 1, $\mathrm{Re}\left(z\overline{z'}\right) \leq \left|z\overline{z'}\right| = |z|\,|z'|$. So $|z+z'|^2 \leq (|z| + |z'|)^2$, and both numbers compared are nonnegative.
4. Equality is equivalent to $\mathrm{Re}\left(z\overline{z'}\right) = \left|z\overline{z'}\right|$, that is, by 1, to $z\overline{z'} = \mu$, a nonnegative real number. In that case, $z = \frac{\mu}{|z'|^2}z'$ with $\lambda = \frac{\mu}{|z'|^2} \geq 0$. Conversely, if $z = \lambda z'$ with $\lambda \geq 0$, $z\overline{z'} = \lambda\,|z'|^2 \geq 0$. Geometrically: M lies on the ray $[OM')$.
5. (a) Let $w = 3+4i$, whose modulus is 5. On the one hand $|z| = |(z-w)+w| \leq 2+5 = 7$. On the other hand $5 = |w| = |(w-z)+z| \leq |z-w| + |z| \leq 2 + |z|$, so $|z| \geq 3$.
   (b) $|z| = 7$ requires equality in $|(z-w)+w| \leq |z-w| + |w|$ and $|z-w| = 2$: $z - w = \lambda w$ with $\lambda \geq 0$ and $|\lambda w| = 2$, that is $\lambda = \frac{2}{5}$ and $z = \frac{7}{5}w = \frac{21}{5} + \frac{28}{5}i$. Likewise, $|z| = 3$ gives $w - z = \frac{2}{5}w$, that is $z = \frac{3}{5}w = \frac{9}{5} + \frac{12}{5}i$. Both points do satisfy $|z-w| = 2$.

Teacher note · Status of the problem. The triangle inequality and its case of equality are a "possible problem" of the syllabus. Question 5 is easy to visualise: the disc with centre $(3,4)$ and radius 2 lies between the circles with centre O and radii 3 and 7, which it touches on the line joining O to the centre of the disc.